Analysis brief PA-3: Breeding-pair dimorphism

Seabird Ecology Group · version 1.2 · applies to the three-season pair study, 2007/08 to 2009/10

1. Purpose and inputs

This brief sets the rules for turning the raw nest records into a table of breeding pairs and a measure of how much heavier males are than their mates. Every figure in the outputs must be traceable to `penguins_raw.csv` and the field corrections log, using only the rules below.

The raw file holds one row per adult. The corrections log is a scan of the typed-up field sheets and has to be read from the page.

2. Records, seasons and nests

- **Season** comes from `studyName`: PAL0708 is 2007/08, PAL0809 is 2008/09 and PAL0910 is 2009/10. Do not derive the season from the egg date.
- **Species** is the first word of the `Species` column: Adelie, Chinstrap or Gentoo. Wherever species are listed or drawn, use that order.
- **Bird ID** is `Individual ID`, written N<nest number>A<adult number>. Bird IDs are reused between seasons, so a record is identified only by season and bird ID together.
- **A nest** is the two adults that share a season and a nest number. Nest-level facts (island, egg date, clutch completion) are the same for both adults.

3. Applying the corrections log

Apply the log to the raw records before any other step. For each entry, decide its status in this order:

- **Void – withdrawn.** The entry has been struck through or marked withdrawn.
- **Void – no matching record.** There is not exactly one record whose season, bird ID and island all match the entry.
- **Void – recorded value does not match.** The entry's "recorded" value disagrees with the raw file for that record. Compare like with like: log dates are day/month/year, masses in the log may be in kilograms while the raw file is in grams, and "not recorded" means the raw value is missing.
- **Superseded.** Two or more otherwise valid entries change the same field of the same record. Only the one with the latest "logged" date is applied; the others are superseded. Entry numbers follow the order the sheets were typed up, not the order they were logged.
- **Applied.** Every other entry. Where the log and the raw file disagree, an applied entry wins.

Void and superseded entries change nothing.

4. Sex and nest status

After corrections, give every nest one status:

- **incomplete** – either adult has no body mass or no flipper length. This holds even if both sexes are known.
- **unresolved** – both adults were measured but neither has a recorded sex.
- **complete** – both adults were measured and at least one has a recorded sex. If only one adult has a recorded sex, its mate takes the opposite sex and the nest's sex source is `inferred`; otherwise the sex source is `recorded`. A sex set by an applied correction counts as recorded. No nest has two adults recorded as the same sex.

5. Pair measures (complete nests only)

- **Mass dimorphism (%)** = (male mass − female mass) ÷ female mass × 100.
- **Flipper difference (mm)** = male flipper length − female flipper length.
- **Reversed** = yes when the male is lighter than the female, otherwise no. Equal masses are not reversed.

For every nest, whatever its status:

- **Lay day** = the egg date counted in days from 1 November of the year the season began, with 1 November as day 1. For 2008/09, 2 November 2008 is day 2.

- **Clutch failed** = yes when `Clutch Completion` is No, otherwise no.

6. Rounding

Do every calculation on unrounded values and round only what is written out. Percentages and means are written to one decimal place, rounding halves away from zero (2.25 becomes 2.3, −2.25 becomes −2.3). Medians and means in the summary are taken over the unrounded pair values.

7. Output files

All outputs are saved on the Desktop with these names. Empty cells are left blank, not written as NA.

pair_table.csv – one row per nest, sorted by season, then species, then nest number.

Column	Content
`season`	2007/08, 2008/09 or 2009/10
`species`	Adelie, Chinstrap or Gentoo
`island`	As in the raw file
`nest`	Nest number as a whole number
`status`	complete, unresolved or incomplete
`sex_source`	recorded or inferred; blank unless complete
`male_id`	Bird ID of the male; blank unless complete
`female_id`	Bird ID of the female; blank unless complete
`male_mass_g`	Blank unless complete
`female_mass_g`	Blank unless complete
`mass_dimorphism_pct`	One decimal place; blank unless complete
`flipper_diff_mm`	Blank unless complete
`reversed`	yes or no; blank unless complete
`lay_day`	Whole number, every nest
`clutch_failed`	yes or no, every nest

species_season_summary.csv – one row per season and species, sorted by season, then species.

Column	Content
`season, species`	As above
`n_nests`	All nests in the group
`n_complete, n_unresolved, n_incomplete`	Nests of each status
`n_failed`	Nests with clutch failed = yes, any status
`median_mass_dimorphism_pct`	Median over complete pairs, one decimal place
`mean_flipper_diff_mm`	Mean over complete pairs, one decimal place
`n_reversed`	Complete pairs with reversed = yes
`median_lay_day`	Median over all nests in the group, one decimal place

corrections_audit.csv – one row per log entry, in entry order, with columns `entry`, `status` (applied, superseded or void) and `reason` (for void entries: withdrawn, no matching record, or recorded value does not match; blank for applied and superseded entries).

pair_summary.md – a short note for the season report giving, as plain statements: the number of nests, complete pairs, unresolved nests, incomplete nests, pairs with inferred sex, reversed pairs and failed nests; the season and species with the highest and with the lowest median mass dimorphism and those values; and how many log entries were applied out of how many. Each figure must agree with the tables.

pair_analysis.R – the script that produces all of the above from `penguins_raw.csv` when run in RStudio from the Desktop. The corrections are entered in the script; it must not need the internet.

pair_dimorphism.png – the figure described in section 8.

8. Figure standard

One PNG, 1600 × 900 pixels, showing mass dimorphism for complete pairs only.

- Three panels side by side, one per species in the standard order, sharing one vertical scale. Each panel is titled with the species name.
- Horizontal axis: the three seasons in date order. Vertical axis: mass dimorphism (%).
- Every complete pair is one marker. Ordinary pairs are filled circles. Reversed pairs are hollow red triangles, so they stand out from the rest.
- A dashed horizontal line marks zero in every panel.
- For each season in each panel, a short solid horizontal bar marks the median, with the median value written next to it to one decimal place.
- Main title: “Within-pair body mass dimorphism by species and season”. Horizontal axis label: “Season”. Vertical axis label: “Male heavier than female (%)”.
- Check the rendered image before saving. No median label may sit on top of a marker, a bar or another label, and no text may be cut off or overlap other text. Move things until it reads cleanly.

Layout sketch (positions only; values are placeholders):

